

LOCALCONNECT

AN NLP ASSISTED BOOKING PLATFORM

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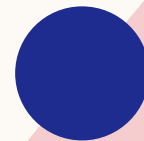
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INTRODUCTION

Local service booking platform is a web application where customers can connect with service providers.

Here in this Application customers can search for services such as :-

- Electrician
- Plumber
- Carpenter
- Ac Repair
- Painter and so on.

This project provides a **single platform** where customers can describe their requirements in natural language and receive suitable service provider recommendations.

Example

“My AC is not cooling. I need a technician today within 5 km and my budget is ₹700.”

The system understands the requirement and recommends suitable providers.

PROBLEM STATEMENT

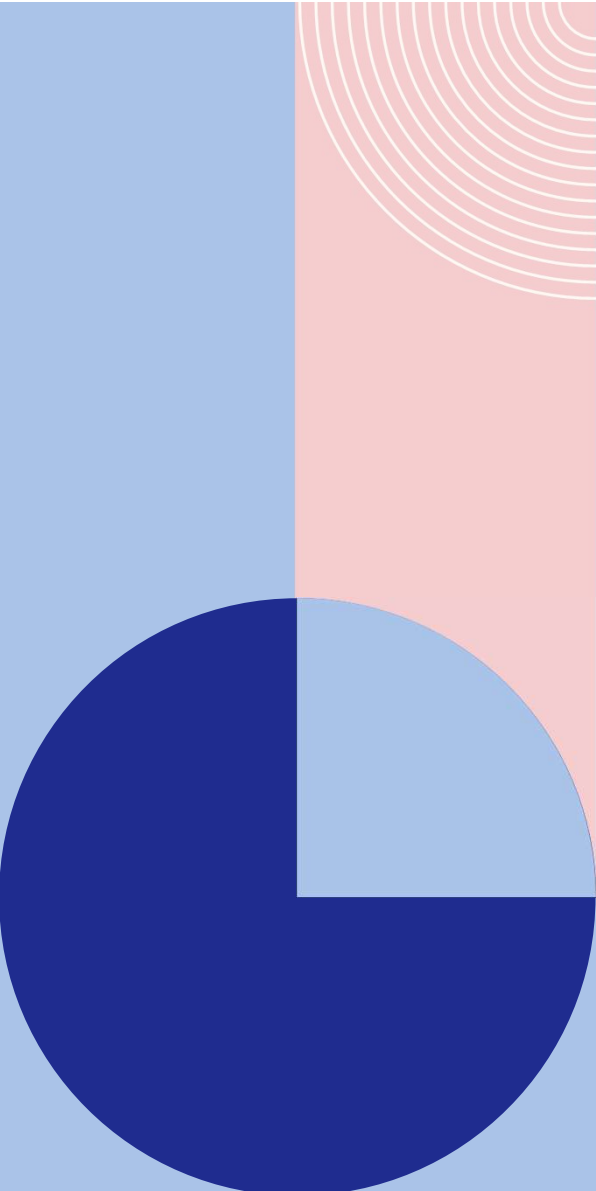
Finding a reliable local service provider can be difficult.

Problems faced by customers:

- It is difficult to find a trusted service provider.
- Customers cannot easily compare different providers.
- Price information may not be clear.
- Provider availability may not be known.
- Traditional booking often requires phone calls.
- Customers may not have a proper way to communicate with providers before booking.
- There is no centralized booking history.

Problems faced by service providers:

- Difficult to find new customers.
- Difficult to manage multiple bookings.
- Communication with customers can be unorganized.
- Limited online visibility.



OBJECTIVES

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- **Easy Service Search** – It helps user to finds services easily.
- **NLP-Based Requirement Understanding** – It understand customer requirement in natural language.
- **Easy Booking** – Users can select a provider, date, and time to book a service.
- **Ratings & Reviews** – Customers can rate and review the service provider.
- **Provider Management** – Providers can manage their services, availability, and bookings.
- **Admin Management** – Admin can manage users, providers, services, and complaints.
- **Personalized Experience**- Use booking history and provider performance to improve recommendations.

EXISTING SYSTEM

Many platforms already provide **local service booking such as--:**

- Urban Company
- NoBroker
- Housejoy

In this system customer selects the required service .system shows available service providers. Customer views provider details. Customer compares providers based on **price, rating, and other details**. Customer selects a provider. Customer books the service.

Limitations of Existing System--:

Customers have to manually compare different providers.

Recommendations are not always personalized.

User requirements written in natural language may not be fully understood.

Factors like location, price, rating, availability, and previous bookings may not be combined effectively

PROPOSED SYSTEM

We propose an **NLP-assisted Local Service Booking** Platform. The customer can describe their requirement in a simple sentence. The customer does not need to search and compare providers manually.

The NLP system understands the customer's requirements.

It identifies important details such as:

- a. Type of service
- b. Location/distance
- c. Required date and time
- d. Budget
- e. Other preferences

The system then finds suitable service providers.--:

- i) Providers are ranked using factors such as rating, price, location, availability, and past bookings.
- ii) The system recommends the most suitable provider to the customer

KEY FEATURES

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1. Natural Language Search

- Customer can describe their problem in simple language.
- No need to search using specific keywords.

Example: “My washing machine is making a strange noise and I need someone tomorrow.”

2. NLP-Based Information Extraction

NLP understands the customer's sentence.
It extracts important information such as:

- Service
- Problem
- Budget
- Distance
- Date/Time

3. Intelligent Recommendation

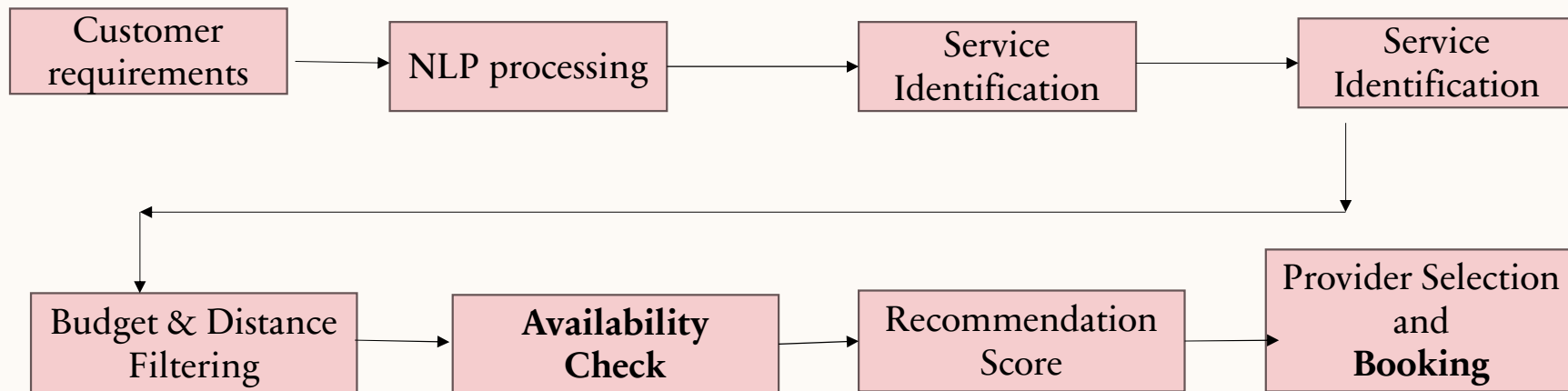
The system finds and ranks the best service providers.
Ranking is based on:

- Rating
- Distance
- Price
- Availability
- Experience
- Reliability
- Booking History



PROPOSED SOLUTION

- Our system combines NLP, Recommendation Algorithm, and Booking System.
- The customer simply describes their requirement in natural language.
- NLP understands the requirement and extracts important details.
- The system identifies the required service, budget, distance, and date.
- It filters providers based on the customer's requirements.
- It checks the availability of service providers.



TECHNOLOGY STACK

Frontend---

- React.js
- JavaScript
- Tailwind CSS
- React Router
- Axios

Backend---

- Node.js
- Express.js
- REST API

Database---

- MongoDB
- Mongoose

NLP & Recommendation---

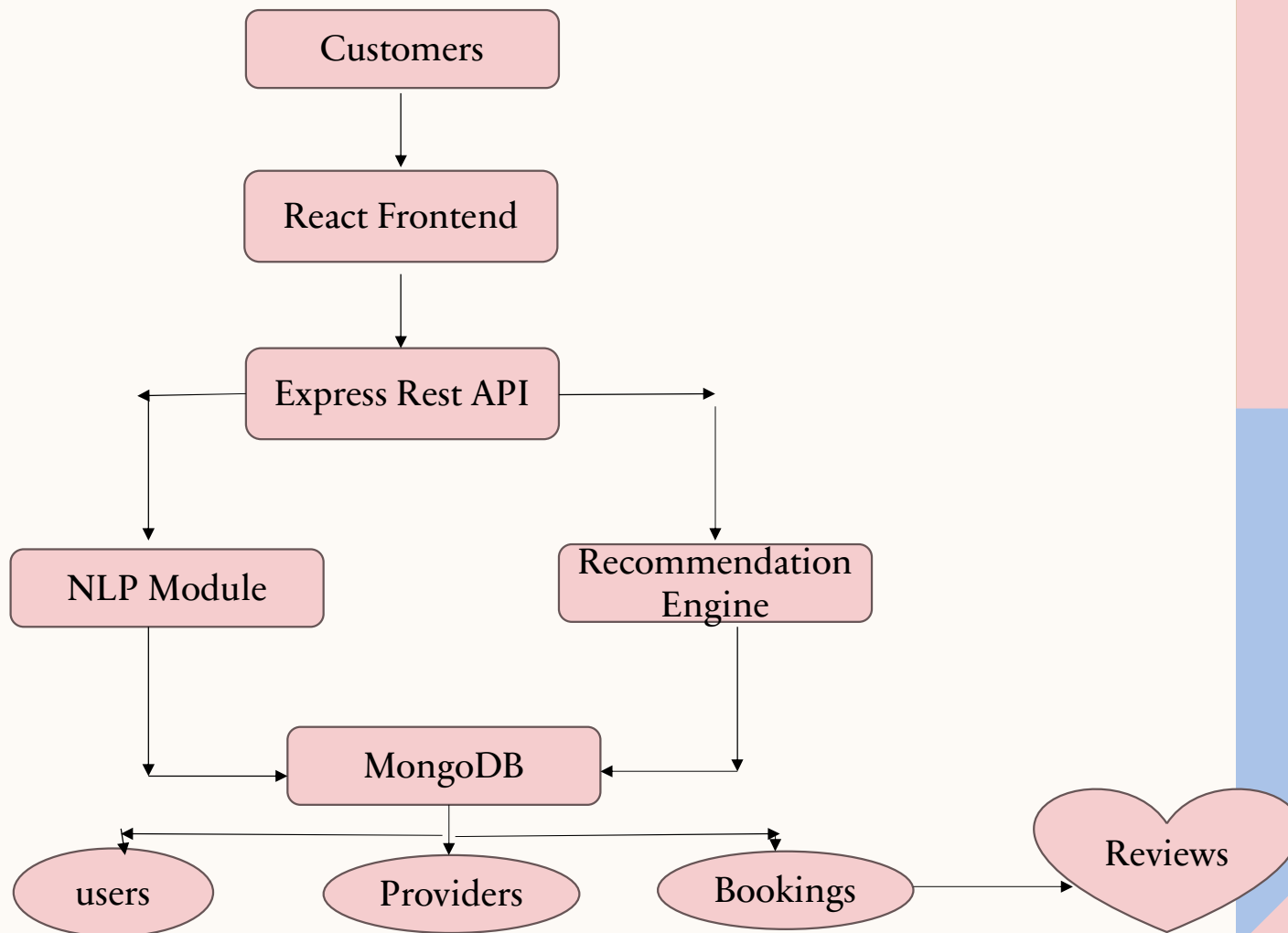
- JavaScript-based NLP
- Keyword matching
- Regular Expressions
- Weighted Recommendation Algorithm
- Haversine Distance Formula

Tools---

- VS Code
- Git
- GitHub
- Postman
- MongoDB Compass

SYSTEM ARCHITECTURE & DATABASE .

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Our system uses **MongoDB** because it provides **flexible and document-based data storage**. -----:

1. User Collection

- Stores information about customers and users.
- **Name** – User's name
- **Email** – User's email address
- **Password** – Login password
- **Phone** – Contact number
- **Role** – Customer or provider
- **Location** – User's location

2. Provider Collection

- Stores information about service providers.
- **Business Name** – Name of the business
- **Services** – Services provided
- **Location** – Provider's location
- **Price** – Service price
- **Rating** – Customer rating
- **Availability** – Available date/time
- **Experience** – Provider's experience
- **Booking Statistics** – Previous booking information

3. Booking Collection

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- Stores details of customer bookings.
- **Customer** – Customer who made the booking
- **Provider** – Selected service provider
- **Service** – Booked service
- **Booking Date** – Date and time of service
- **Address** – Service location
- **Price** – Booking cost
- **Status** – Pending, confirmed, completed, or cancelled

4. Review Collection

- Stores customer feedback about providers.
- **Customer** – Person giving the review
- **Provider** – Provider being reviewed
- **Booking** – Related booking
- **Rating** – Rating given by customer
- **Comment** – Customer's feedback

FUTURE SCOPE & CONCLUSION

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Our current system can be extended with more advanced technologies.

- **Advanced AI/LLM**
- **Real-Time GPS**
- **Smart Availability**
- **Online Payment**
- **Real-Time Chat**
- **Notification**
- **Machine Learning**

Our Local Service Booking Platform with Recommendation provides a smarter approach to local service discovery and booking.

Our system:

Understands customer requirements using NLP.

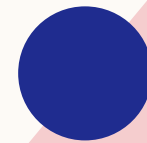
Extracts service, problem, budget, distance, and date.

Filters providers based on customer requirements.

Provides explainable recommendations.

Allows customers to book services.

Stores booking and review information.



**THANK
YOU**